SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA04**

COURSE OUTCOME COVERED

CO1: *Analyse the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL4)*

QUESTIONS: 05 **Total Marks:100**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I ***_Monika.R(192324281)_*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Monika R

Annexure A

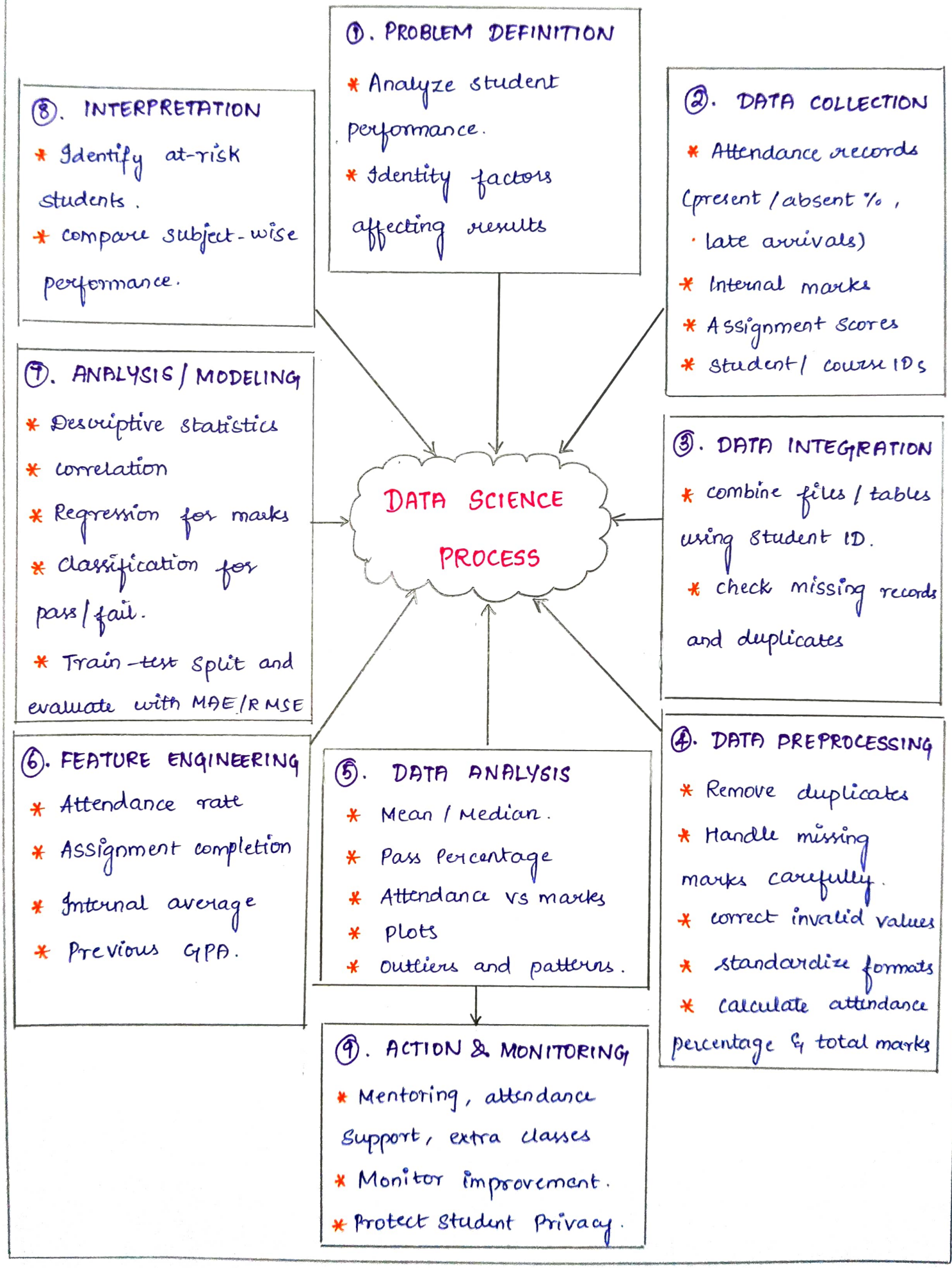
Concept Mapping

1. A college wants to analyze student performance using attendance records, internal marks, assignment scores, and final exam results. Draw a concept map showing the Data Science process from data collection to data transformation.
2. A hospital collects patient details from registration forms, lab reports, medical devices, and doctor notes. Construct a concept map showing the different data collection sources and how they are used for analysis.
3. An online shopping company finds missing customer details, duplicate orders, wrong product entries, and unusual purchase amounts in its dataset. Draw a concept map showing the data cleaning steps required before analysis.
4. A smart city project collects traffic data from CCTV cameras, GPS devices, sensors, and mobile applications. Prepare a concept map showing how data from multiple sources can be integrated into a single dataset.
5. A bank wants to prepare customer data for loan approval prediction. The dataset contains income, age, job type, loan amount, and credit score. Draw a concept map showing suitable data transformation techniques such as encoding, normalization, and feature scaling.

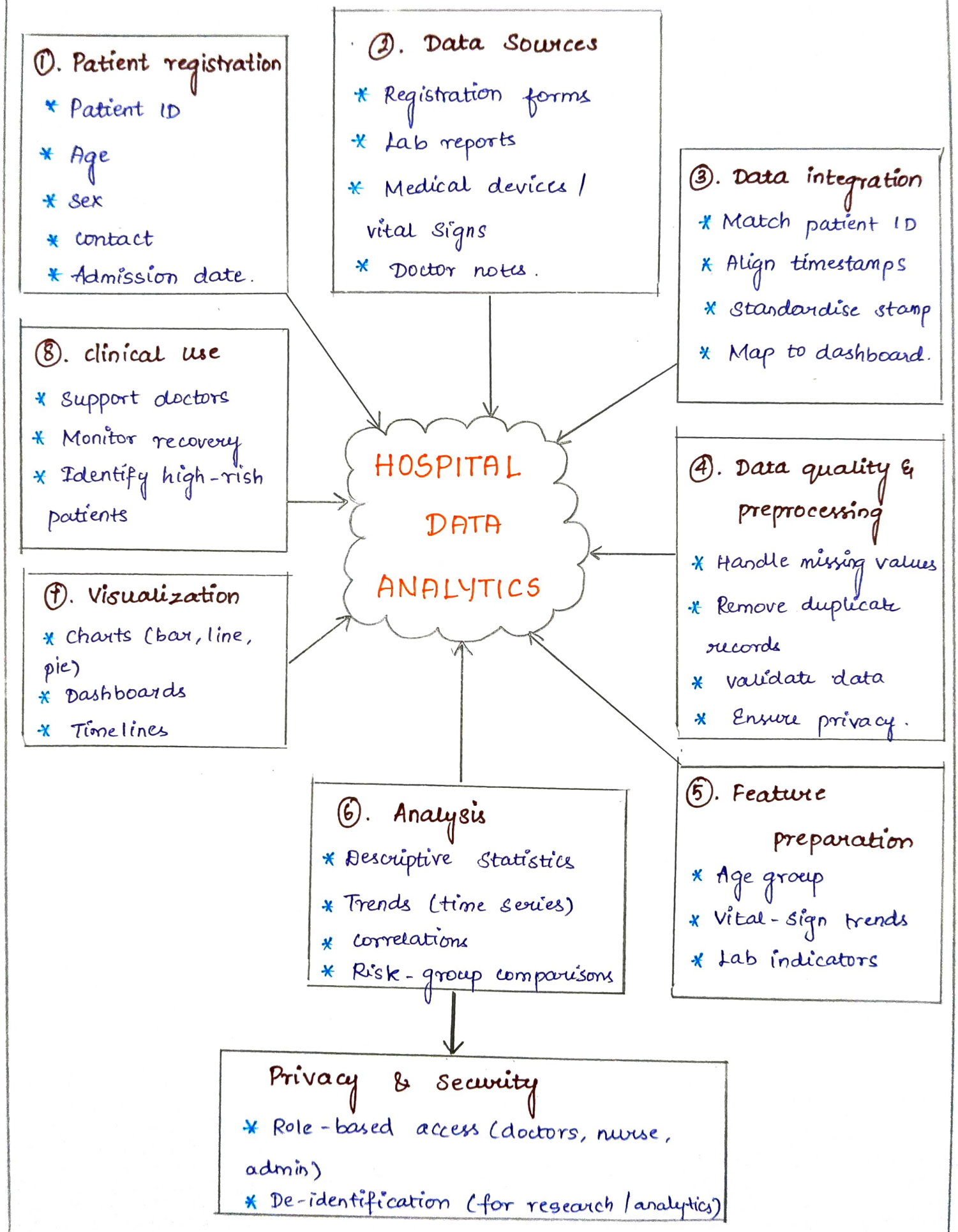
RUBRICS FOR CALCULATION

Concept Mapping					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

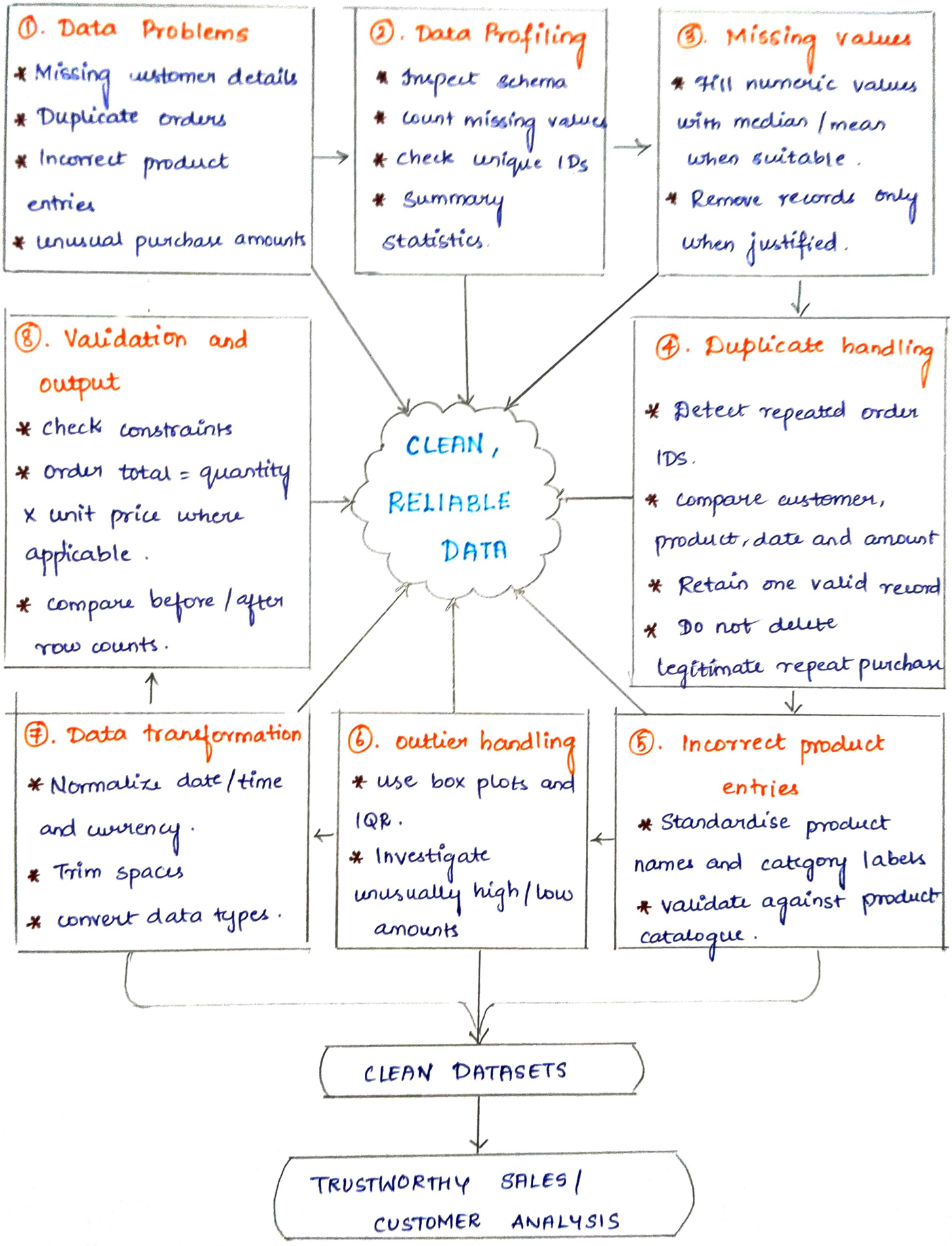
Q1. STUDENT PERFORMANCE ANALYSIS



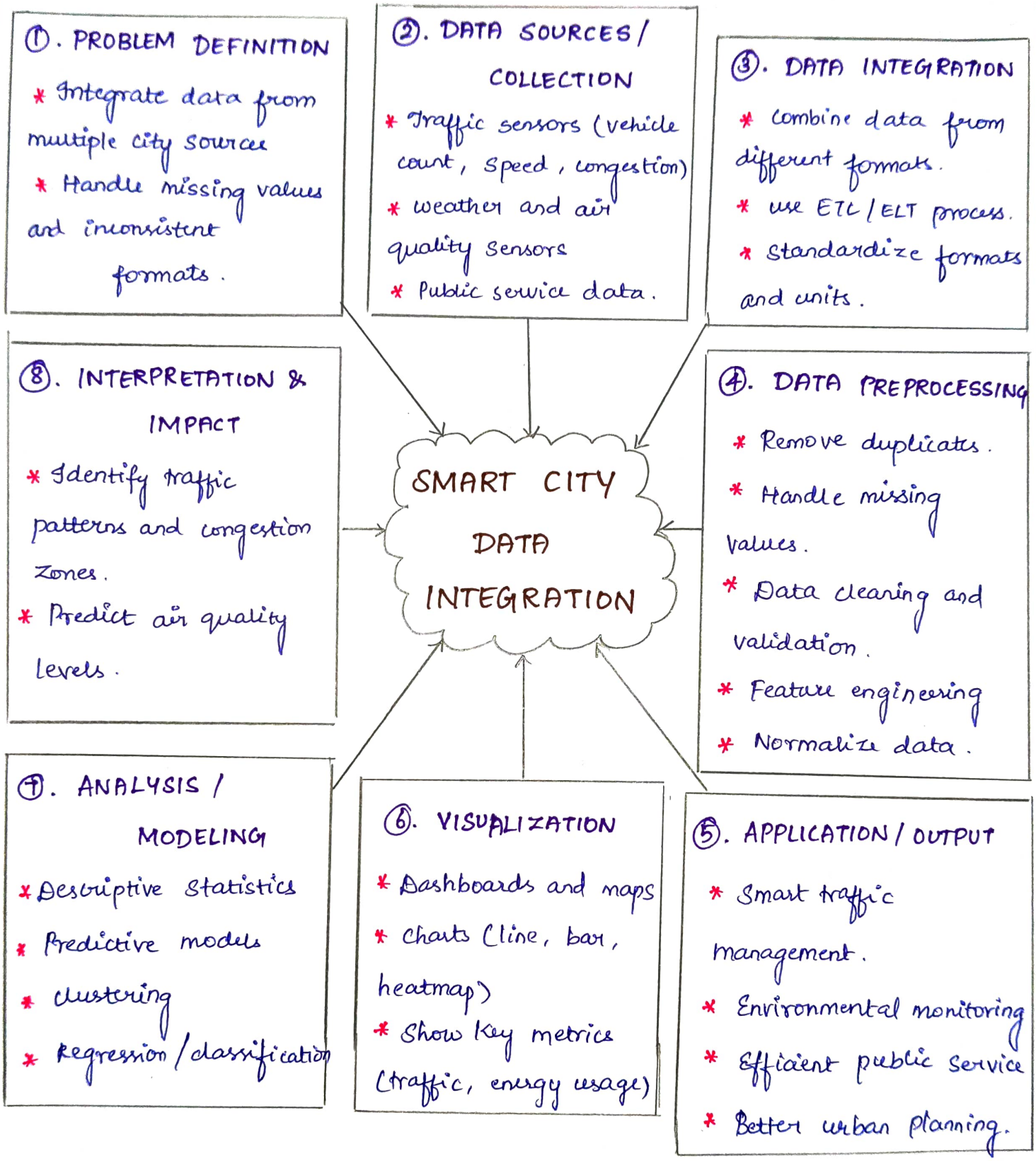
Q2. HOSPITAL PATIENT DATA ANALYSIS



Q3. ONLINE SHOPPING DATA CLEANING



Q4. SMART CITY DATA INTEGRATION



①. PROBLEM DEFINITION

* Integrate data from multiple city sources
* Handle missing values and inconsistent formats.

②. DATA SOURCES / COLLECTION

* Traffic sensors (vehicle count, speed, congestion)
* weather and air quality sensors
* Public service data.

③. DATA INTEGRATION

* Combine data from different formats.
* use ETL / ELT process.
* Standardize formats and units.

⑧. INTERPRETATION & IMPACT

* Identify traffic patterns and congestion zones.
* Predict air quality levels.

④. DATA PREPROCESSING

* Remove duplicates.
* Handle missing values.
* Data clearing and validation.
* Feature engineering
* Normalize data.

⑦. ANALYSIS / MODELING

* Descriptive statistics
* Predictive models
* clustering
* Regression / classification

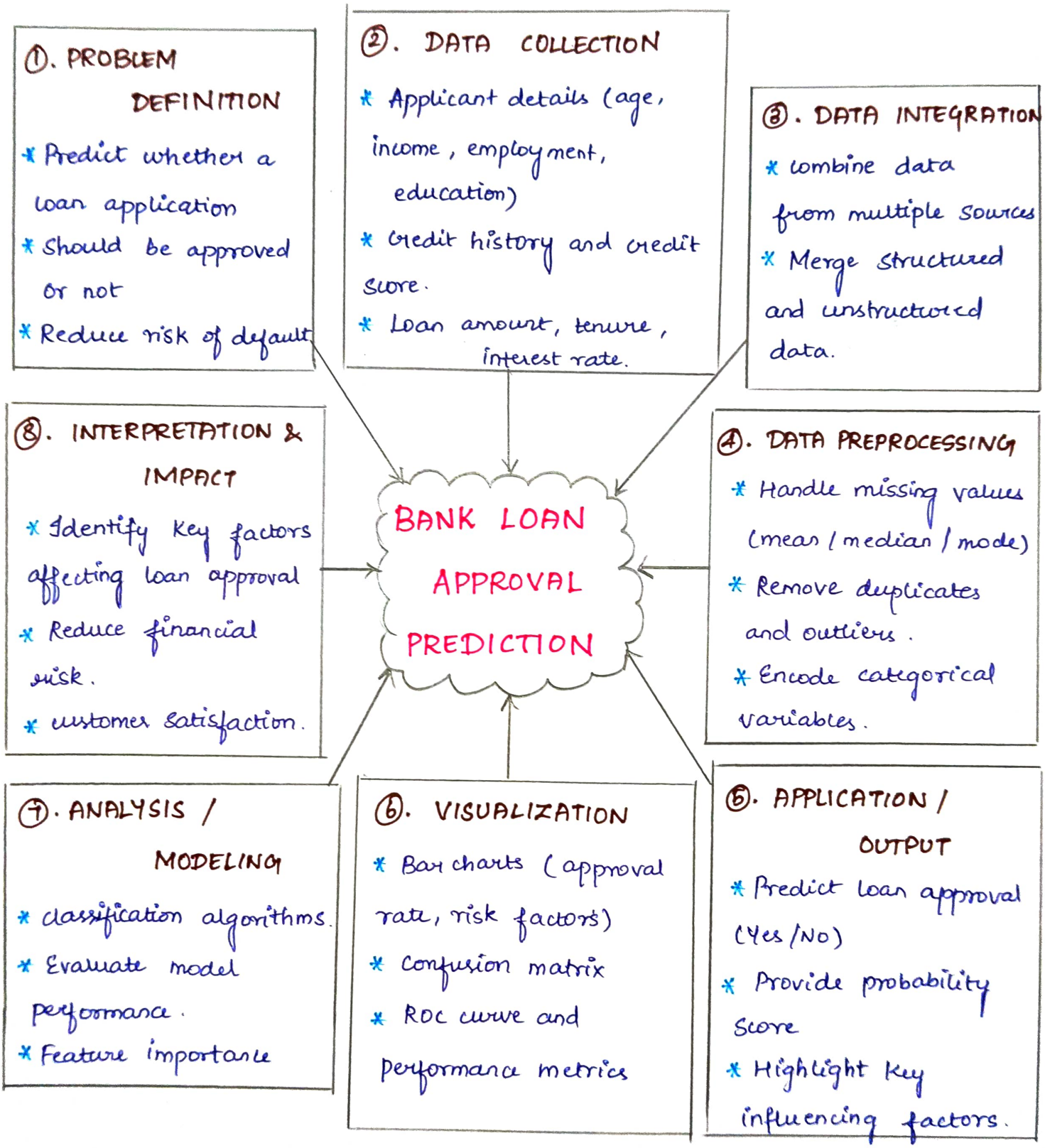
⑥. VISUALIZATION

* Dashboards and maps
* Charts (line, bar, heatmap)
* Show key metrics (traffic, energy usage)

⑤. APPLICATION / OUTPUT

* Smart traffic management.
* Environmental monitoring
* Efficient public service
* Better urban planning.

Q5. BANK LOAN APPROVAL PREDICTION



①. PROBLEM DEFINITION

* Predict whether a loan application
* Should be approved or not
* Reduce risk of default

②. DATA COLLECTION

* Applicant details (age, income, employment, education)
* Credit history and credit score.
* Loan amount, tenure, interest rate.

③. DATA INTEGRATION

* Combine data from multiple sources
* Merge structured and unstructured data.

④. DATA PREPROCESSING

* Handle missing values (mean / median / mode)
* Remove duplicates and outliers.
* Encode categorical variables.

⑤. APPLICATION / OUTPUT

* Predict loan approval (Yes / No)
* Provide probability score
* Highlight key influencing factors.

⑥. VISUALIZATION

* Bar charts (approval rate, risk factors)
* Confusion matrix
* ROC curve and performance metrics

⑦. ANALYSIS / MODELING

* classification algorithms.
* Evaluate model performance.
* Feature importance

⑧. INTERPRETATION & IMPACT

* Identify key factors affecting loan approval
* Reduce financial risk.
* customer satisfaction.